Understanding ACT: Loss and training

Explaining the Style Variable Generation

1 The Dual Loss Objective

The complete training of ACT relies on two mathematical penalties working in tandem. The model must simultaneously become a precise physical imitator and an efficient style summarizer. The objective function evaluated during each step is defined as:

$$\mathcal{L}_{total} = \mathcal{L}_{reconst} + \beta \mathcal{L}_{KL}$$

1.1 L1 Reconstruction Loss ($\mathcal{L}_{reconst}$)

The primary goal of the Policy Decoder is to execute actions that match the human expert's original demonstration. While deep learning often relies on Mean Squared Error (L2 Loss) to measure differences, the ACT authors chose a variant of the **L1 Loss**.

Technically, pure L1 loss ($|x|$) is non-differentiable at exactly zero. To solve this, implementations usually use **SmoothL1Loss** (also known as Huber Loss). This function is quadratic when the error is very small (making it smooth and differentiable at zero) and linear when the error is large.

$$\mathcal{L}_{reconst} = \text{SmoothL1}(\hat{a}, a)$$

L1 Loss is deliberately chosen over L2 because L2 heavily penalizes large outliers but is very forgiving of tiny errors. In physical robotics, being "slightly off" across many joints leads to catastrophic, "blurry" movements. L1 Loss forces much tighter adherence to the precise, fine-grained details required for manipulation.

1.2 KL Divergence ($\mathcal{L}_{KL}$) and The Multi-Modal Averaging Problem

One might argue: if we just set $z = \vec{0}$ at inference time anyway, why do we bother having a style variable z during training? Why not just train a single Encoder-Decoder without the CVAE and force it to learn one single way to do the task?

The answer is the **Multi-Modal Averaging Problem**. Human demonstrations are messy and "multi-modal." Given the exact same starting position, a human might execute a task in two different ways (e.g., picking up a cup by the handle vs. from the top).

- **Without z:** A standard neural network, seeing two conflicting actions for the same image, will try to minimize the L1 Loss by *averaging* the two actions. It will steer right down the middle and crash into the side of the cup.
- **With z:** By introducing the style variable z, we give the model an "excuse" for the conflicting data. It learns that when z is positive, it should grab the handle, and when z is negative, it should grab the top. It successfully learns both methods without improperly averaging them.

1.2.1 Why the Standard Normal Prior $\mathcal{N}(0, I)$ is Required

If we allow the model to use z to separate these different "modes" (styles) during training, we still face a problem at inference: the human is no longer there to demonstrate, meaning the CVAE Encoder cannot run. We have to "guess" a z vector out of thin air.

If we did not use the KL Divergence penalty, the model would spread these different styles randomly across the infinite 512-dimensional space. Guessing a random coordinate would result in erratic, untested behavior.

By enforcing a massive KL Divergence penalty ($\beta = 10$), we violently force the model to cluster all of these different human modes tightly around the origin coordinate $(0, 0, \ldots, 0)$. This mathematical constraint gives us a profound advantage for inference:

- Because we forced all valid styles to cluster around zero, setting $z = \vec{0}$ at inference time acts as a clean "tie-breaker."
- It guarantees that the Policy Decoder will execute the most robust, dominant, and "safest" average of the strategies it learned, without executing conflicting physical movements.

2 Optimization Procedure

The entire architecture—including both the CVAE Encoder (parameters ϕ) and the Policy Transformer (parameters θ)—is optimized simultaneously in a single backward pass. By combining both loss constraints, the gradients flow backward from the Decoder's action outputs, through the sample vector z, and all the way back up to the Encoder's attention layers. This forces the model to agree on a style language that is both compact enough to satisfy the KL constraint and descriptive enough to allow flawless physical reconstruction.